

Program 5

Objective: Demonstrate Continuous Integration (CI) by running Jenkins inside a Docker container.

Project Structure

```
lab5/
|
├── Dockerfile
├── docker-compose.yml
└── lab-5.png
```

Source Code

Dockerfile

```
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && apt-get install -y docker.io

USER jenkins
```

docker-compose.yml

```
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
```

- jenkins_home:/var/jenkins_home
- /var/run/docker.sock:/var/run/docker.sock

volumes:
jenkins_home:

Setup and Execution

Create Project Folder

```
1rv24mc078_prateek_kumar@arch mkdir lab5  
1rv24mc078_prateek_kumar@arch cd lab5
```

Build Jenkins Image (Using Dockerfile)

```
1rv24mc078_prateek_kumar@arch ~/lab5  
sudo docker build -t lab5 .
```

Run Jenkins Container

```
1rv24mc078_prateek_kumar@arch ~/lab5  
sudo docker run -p 8080:8080 -p 50000:50000 --name jenkins_ci lab5
```

Using Docker Compose

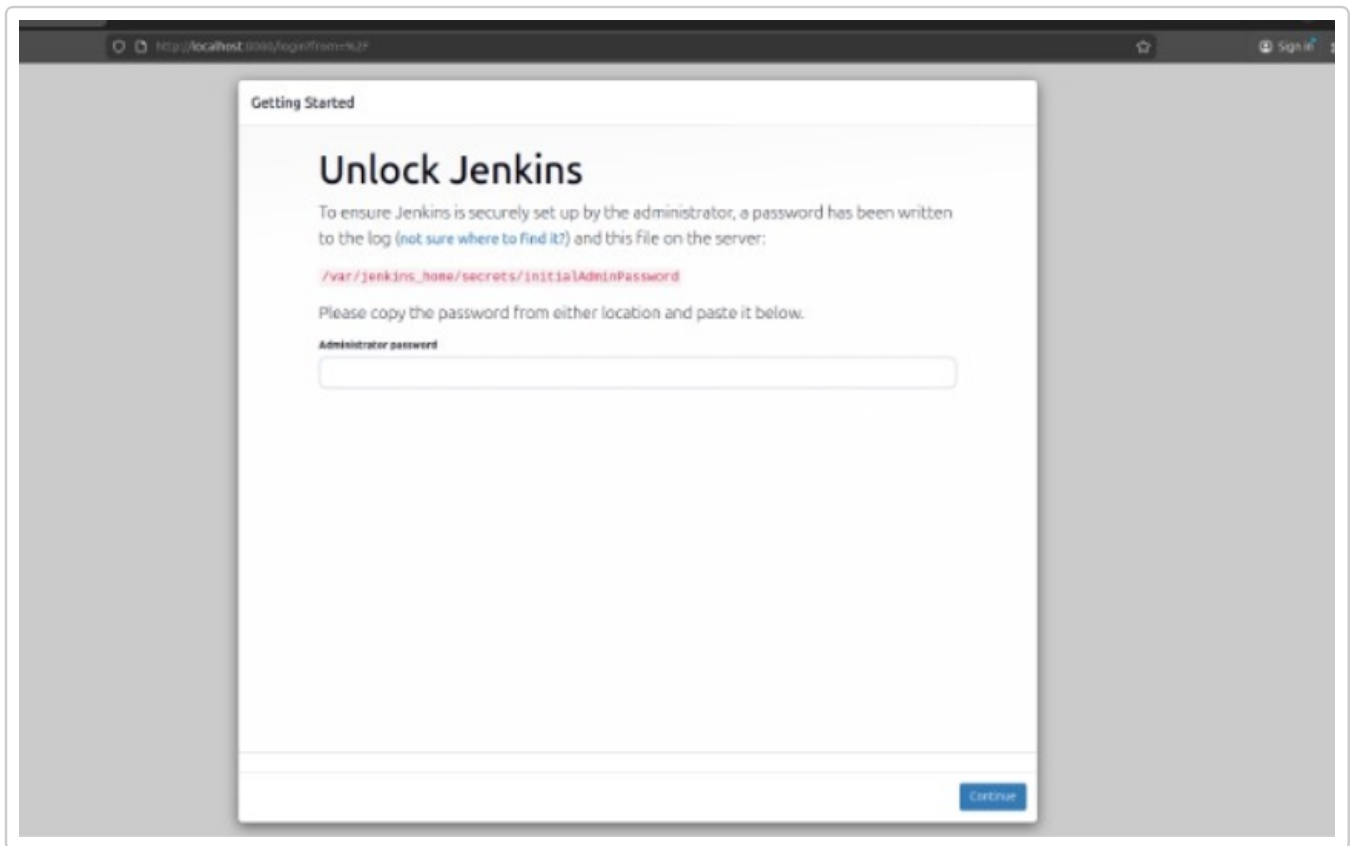
```
1rv24mc078_prateek_kumar@arch ~/lab5  
sudo docker compose up
```

Access Jenkins

<http://localhost:8080>

Output Screenshot

Jenkins Unlock Page (Running in Docker Container):



Result

Jenkins was successfully deployed inside a Docker container, demonstrating CI integration. The Jenkins dashboard was accessed via browser on port 8080.